HARSHIT JAIN

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EXPERIENCE

Data Scientist, Jan 2026 - Present
Indiana University *Bloomington, IN*

- Deployed and compared three photo-age prediction models - DEX VGG-16, MiVOLO transformer, and ArcFace Ridge Regression - across 1,261 subjects, identifying MiVOLO as the strongest performer and improving model accuracy.
- Implemented residualization analysis showing that raw photo-to-epigenetic-clock correlations were substantially inflated by shared age structure, isolating the genuine biological aging signal - collaborated with teammates to cross-validate findings against their independent analysis pipeline.
- Built model interpretability workflows using Grad-CAM activation mapping and a 9-region facial ablation study, finding that the jawline region drives the strongest biological aging signal and that model attention systematically shifts with subject age.
- Applied Lasso regression on 26 geometric facial features extracted via MediaPipe, identifying jaw asymmetry and under-eye hollowing as the primary markers of biological age acceleration - demonstrating that CNN models implicitly encode interpretable facial geometry.

Data Scientist - Research, Jun 2025 - Present
Indiana University *Bloomington, IN*

- Architected a large-scale data collection and vulnerability scanning pipeline on IU's HPC cluster, processing 2.6M+ HuggingFace model cards and running security scans across 22,135 GitHub repositories - surfacing 707,985 vulnerabilities including 118,815 high-severity findings.
- Developed a blast radius metric quantifying downstream model exposure from vulnerable repositories, with findings showing up to 83,706 models at risk from a single repository; built a MySQL analytics layer enabling sub-second query response across the full dataset.
- Built and deployed a three-page interactive vulnerability monitoring dashboard (Dash/Plotly) on IU's Jetstream2 research infrastructure, with multi-filter exploration and a fallback mode for offline access.
- Authored a research paper submitted to PEARC 2026 ACM Visualization Showcase, incorporating PI feedback to restructure findings for a research audience and coordinating submission across the team.

Machine Learning Engineer - Remote, Jun 2023 - Aug 2023
Feynn Labs *Mumbai, India*

- Analyzed 100K+ customer records using SQL to find the behavioral patterns and revenue drivers the product and ops teams were asking about.
- Delivered automated Power BI KPI dashboards replacing manual reporting; ran hypothesis-driven A/B analyses to validate business assumptions and quantify anomaly root causes.
- Modeled customer segments using classification and clustering in Scikit-learn, identifying 4+ distinct cohorts that directly shaped product positioning decisions.
- Consolidated disparate operational datasets into a unified schema using Python and SQL, cutting ad hoc query turnaround by ~35%.

PROJECTS

LLM-Powered Model Search & Recommendation System - *RAG, LoRA, LLM Evaluation* | Link

- Built a model recommendation system over 5000 Hugging Face model cards - combined a RAG pipeline (ChromaDB + MiniLM) with a LoRA fine-tuned TinyLlama-1.1B and a ragas eval layer; RAG outperformed fine-tuning by 18% on faithfulness when training data was limited.

Security Threat Intelligence Platform - *Spark, Postgres, MLflow* | Link

- Architected a distributed Spark ML pipeline ingesting 257K+ network security events into structured analytics tables, training multi-class attack classifiers to 84% AUC - all experiments tracked in MLflow.

Cardiovascular Health Analysis - *BigQuery, Looker Studio* | Link

- Engineered a cloud analytics pipeline across 68K+ patient records, built cohort-level risk features, and the dashboards ended up being used by the clinical team to flag high-risk patients.

SKILLS

Languages: Python, R, PySpark, PostgreSQL
Machine Learning: TensorFlow, PyTorch, Deep Learning, Computer Vision, Feature Engineering, Hyperparameter Tuning, Model Deployment
Generative AI: Large Language Models , Prompt Engineering, NLP, LlamaIndex
Analytics: Statistical Inference, Hypothesis Testing
Tools: Docker, AWS (S3, EC2, SageMaker, Lambda), Tableau, Git

EDUCATION

Indiana University Bloomington | Master of Science in Data Science Aug 2024 - May 2026
University of Mumbai | Bachelor of Technology in Electronics & Telecommunication Engineering Dec 2020 - Jun 2024